

Clinical assessment and vision loss

Start with timing and laterality, localise the loss, then choose the next action.

The first clinical assessment

Assess in a consistent order

- 1** History: onset and duration; one or both eyes; pain, redness, flashes, floaters, curtain, distortion or diplopia. Ask about trauma, surgery, contact lenses and systemic symptoms.
- 2** Measure distance acuity in each eye with usual correction, then pinhole when reduced. Record counting fingers, hand movements or light perception if needed.
- 3** Compare pupils and test for a relative afferent pupillary defect (RAPD). Check colour, confrontation fields, eye movements and lid position.
- 4** Examine the anterior segment, then the fundus. Measure pressure when safe. A suspected open globe needs a shield and urgent review without pressure on the eye.

Clues that change urgency

Presentation	Immediate clinical priority
Chemical exposure	Irrigate immediately, then assess.
Sudden persistent loss	Emergency eye assessment; consider retinal stroke and giant cell arteritis.
Red eye with reduced vision	Exclude corneal infection, uveitis, acute angle closure and postoperative infection.
Flashes or a new curtain	Urgent dilated peripheral retinal examination.

Case presentation

State the eye, onset, pain, measured acuity, pupils and key positive findings. For example: sudden painless left monocular loss with a left RAPD and retinal whitening.

Localise visual loss

Use the time course

Pattern	Leading causes and discriminating clue
Sudden painless monocular loss	Retinal artery occlusion, retinal vein occlusion, retinal detachment or ischaemic optic neuropathy. Fundus and pupil findings separate them.
Transient monocular blackout	Retinal ischaemia is a TIA until assessed; a normal examination after recovery does not exclude it.
Pain on eye movement	Optic neuritis when colour and central vision are reduced, often with RAPD.
Painful red eye	Corneal disease, uveitis or acute angle closure. Check corneal staining, anterior chamber and pressure when safe.
Gradual central distortion	Macular disease; use an Amsler grid and macular OCT.
Gradual glare or peripheral loss	Lens opacity causes glare; glaucoma often causes peripheral field loss before central acuity declines.

A normal fundus does not end the assessment

Retrobulbar optic neuritis, early retinal ischaemia and visual pathway disease may have little visible fundus change. Interpret symptoms, colour, fields and pupils together.

Both eyes affected

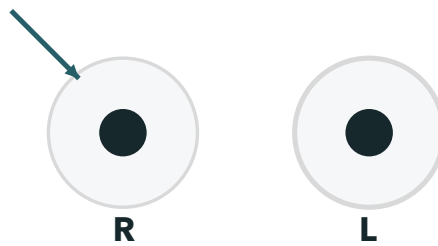
Homonymous field loss localises behind the chiasm. Bitemporal loss suggests a chiasmal process; acute headache, diplopia or hypotension raises concern for pituitary apoplexy.

Bedside localisation

A left RAPD on swinging light test

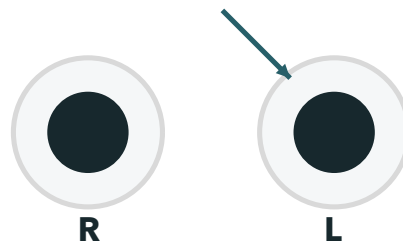
Light in RIGHT eye

Both constrict
Strong afferent signal



Light moved to LEFT eye

Both relax relatively
Weaker left afferent signal



Schematic pupil responses with intact efferent pathways. Apparent dilation reflects a weaker afferent light signal.

Link the finding to function

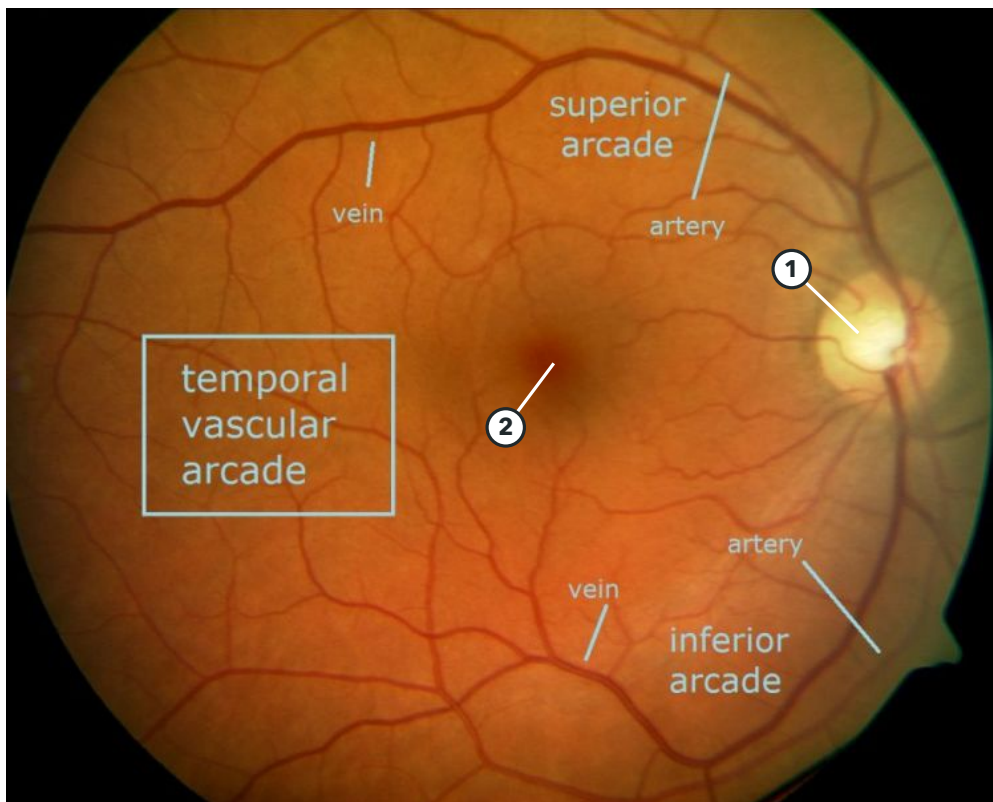
Test	Useful interpretation
Pinhole	Clear improvement supports an optical or refractive component; it does not exclude coexisting disease.
Red comparison	A red target appears less vivid through the affected eye in many optic neuropathies.
RAPD	Asymmetric optic nerve or substantial retinal dysfunction. Equal pupil size does not exclude it.
Confrontation fields	Screens for gross defects; formal perimetry quantifies them.

Clinical trap

Reduced central acuity and a normal fundus with painful eye movement, red desaturation and RAPD strongly suggest optic nerve disease. Bilateral symmetric disease can lack a RAPD.

Read a fundus photograph systematically

Normal retinal landmarks



1 Optic disc. 2 Foveal centre within the macula. Existing labels identify vessels and arcades. Supplied M3 Anki, Normal Retinal Anatomy, media 162.

Describe before diagnosing

Region	What to assess
Disc	Margins, colour, rim and cup; distinguish swelling from cupping.
Vessels	Calibre, tortuosity, emboli, venous beading and new vessels.
Macula	Haemorrhage, hard exudates, drusen and signs of fluid.
Background	Distribution of haemorrhages, cotton wool spots and peripheral abnormalities.

Name the limitation

A posterior pole photograph cannot exclude a peripheral tear. OCT demonstrates retinal thickness and fluid; angiography assesses perfusion or leakage when indicated.

References AAO fundus interpretation

Choose the next investigation

Ask what the test will answer

Clinical suspicion	Next useful assessment
Macular oedema or wet AMD	Macular OCT; angiography when the diagnosis or leakage needs clarification.
Glaucoma	Gonioscopy, optic nerve assessment, fields and OCT nerve fibre layer; pressure alone is insufficient.
Retinal tear or detachment	Dilated peripheral retinal examination; B-scan if the fundus view is obscured and the globe is intact.
Retinal TIA	Urgent stroke/TIA pathway with vascular and cardiac assessment; recovery is not reassurance.
Giant cell arteritis	ESR, CRP and full blood count, followed by confirmatory vascular testing without delaying steroids.
Papilloedema	Urgent brain imaging plus venography before a lumbar puncture when safe.

Case 1

A 76-year-old with new scalp tenderness, jaw claudication and sudden loss has threatened vision from possible giant cell arteritis. Start urgent systemic glucocorticoids and involve specialists; confirmation must not delay treatment.

Case 2

A 65-year-old describes a dark monocular curtain lasting four minutes, with complete recovery. Arrange an urgent retinal TIA pathway and early secondary prevention through the treating team.